



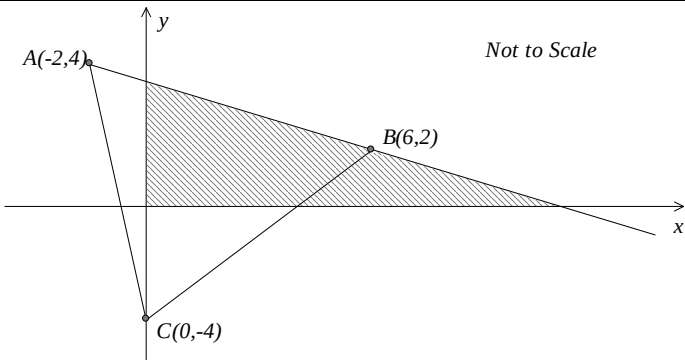
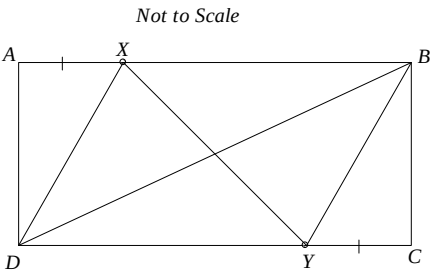
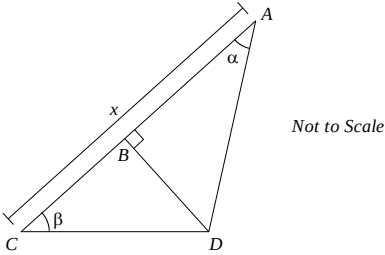
Year 11 Mathematics

June 2010

Time: 65 Minutes

- DIRECTIONS**
- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
 - Use black or blue pen only (*not pencils*) to write your solutions.
 - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.

1.	Find the exact value of a) $\cos 225^\circ$ b) $\cot (-120^\circ)$	1 1
2.	Solve for $0 \leq \theta \leq 360^\circ$ $2\cos\theta + 1 = 0$	2
3.	a) i) Find the value of x°, (no reason required) <i>Not to Scale</i> Given : $AB \parallel EC \parallel DF$ and $EC = CD$ ii) If $CD = a$, $DE = 5$, $BC = 3$ and $AE = a + 1$, find the exact value of a b) <i>Not to Scale</i> $ABDE$ is a rhombus BCD is an equilateral triangle $\angle BAE = 110^\circ$ Find without reasons, the value of BCE	1 3 1
4.	The ratio of the interior to the exterior angles of a regular polygon is 4 : 1 a) Find the size of each exterior angle b) How many sides does the polygon have?	1 1
5.	If $\tan \theta = \frac{2}{3}$ and $\sin \theta < 0$, find the exact value of $\cos \theta$	2

6.	 Not to Scale a) Find the gradient of AB 1 b) Find the distance of AB 1 c) Show the equation of the line AB is $x + 4y - 14 = 0$ 1 d) Find the perpendicular distance from C to the line AB 1 e) Find the point D such that $ADBC$ is a parallelogram 1 f) Find the area of the parallelogram $ADBC$ 1 g) Using inequalities, describe the shaded region above 1
7.	Joe walks 6 kms due east then 8 km on a bearing of 142°. 3 Draw a diagram representing this information and find how far Joe is from his starting point
8.	Not to Scale  $ABCD$ is a rectangle X and Y lie on AB and CD respectively such that $AX = CY$ a) Prove $\triangle AXD \equiv \triangle BCY$ 3 b) Prove that BD and XY bisect each other 2
9.	Hayden and Harry leave a point A. Hayden travels northwest and Harry travels southeast. Hayden and Harry are then 20 kms apart. a) If Hayden travelled twice as far as Harry, how far did Harry walk? 2 b) What is the bearing of Harry's position from Hayden's? 2
10.	a) Prove $\frac{1 + \tan^2 x}{1 + \cot^2 x} = \tan^2 x$ 2 b) Hence solve $\frac{1 + \tan^2 x}{1 + \cot^2 x} = 2 \tan x$ for $0 \leq x \leq 360^\circ$ 3
11.	a) On the same set of axes, sketch the graphs of $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 360^\circ$ 2 b) Hence, find the values of x in this domain for which $\sin x \geq \cos x$ 2
12.	For $0 \leq x \leq 360^\circ$ solve $2 \tan^2 \theta - 5 \sec \theta + 5 = 0$ 4
13.	 Not to Scale $AC = x$ and $DB \perp AC$ Prove that $BD = \frac{x \sin \alpha \sin \beta}{\sin(\alpha + \beta)}$ 3
~ END OF EXAM ~	

$$1. a) \cos 225^\circ = -\cos 45^\circ \quad (47)$$

$$= -\frac{1}{\sqrt{2}} \quad (1)$$

$$b) \cot(-120^\circ) = \cot 240^\circ \\ = \cot 60^\circ \\ = \frac{1}{\sqrt{3}} \quad (1)$$

$$2. \quad 2\cos\theta + 1 = 0 \\ \cos\theta = -\frac{1}{2} \\ \theta = 180-60^\circ, 180+60^\circ \\ = 120^\circ, 240^\circ \quad (2)$$

$$3 a) i) \quad x = 55^\circ \quad (1)$$

$$ii) \quad \left. \begin{array}{l} \frac{a}{3} = \frac{5}{a+1} \\ a^2 + a - 15 = 0 \end{array} \right\} \quad (1) \\ a = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot -15}}{2} \quad (1) \\ = \frac{-1 \pm \sqrt{61}}{2} \\ a > 0 \therefore a = \frac{-1 + \sqrt{61}}{2} \quad (1)$$

$$b) \quad x = 55^\circ$$

$$4. \quad \text{Exterior } \angle = 180 - 156 \\ = 24^\circ$$

$$\therefore \text{No of sides} \\ = 360 \div 24 \\ \therefore \text{No of sides} = 15 \quad (1)$$

$$5 a) \quad m_{AB} = \frac{4-2}{-2-6} \\ = -\frac{1}{4} \quad (1)$$

$$b) \quad d = \sqrt{(4-2)^2 + (-2-6)^2} \\ = \sqrt{4+64} \\ = 2\sqrt{17} \quad (\text{accept } \sqrt{68}) \quad (1)$$

$$c) \quad y-2 = -\frac{1}{4}(x-6) \quad (1) \\ 4y-8 = -x+6 \\ x+4y-14=0$$

$$d) \quad d = |1(0) + 4(-4) - 14|$$

$$e) \quad \frac{10}{\sqrt{17}} \quad (1) = \frac{30}{\sqrt{17}} \quad (1)$$

$$f) \quad \text{Area} = 2 \left(\frac{1}{2} \times \frac{30}{\sqrt{17}} \cdot 2\sqrt{17} \right) \\ = 60 \text{ units}^2 \quad (1)$$

$$g) \quad x > 0, y > 0 \text{ and} \\ x+4y-14 \leq 0 \quad (1)$$

$$5. \quad \cos\theta = -\frac{3}{\sqrt{13}} \quad (1)$$

$$7. \quad a^2 = 6^2 + 8^2 - 2 \cdot 6 \cdot 8 \cdot \cos 128^\circ \\ = 12.9 \quad (1 \text{ d.p.}) \quad (1)$$

$$8. \quad \begin{array}{l} A \quad \begin{array}{|c|c|} \hline X & B \\ \hline \end{array} \\ a) \quad \begin{array}{|c|c|} \hline D & Y+C \\ \hline \end{array} \end{array}$$

(AD = BC (opp. sides of a rect.)
 (1) AX = YC give:
 (1) $\angle A = \angle C = 90^\circ$ (Angles in a rectangle)
 (1) $\triangle AXD \equiv \triangle BYC$ (SAS).

b) $AB = CD$ (opp. sides of a rect.)

Given $AX = YC$

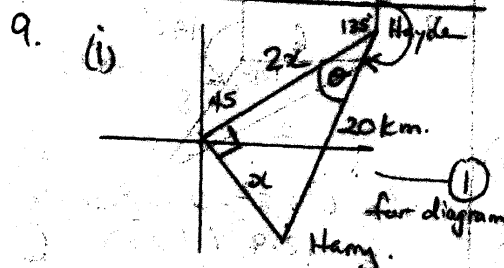
$$BX = DY$$

Since $BX \parallel DY$ (on the opp. sides of a rect.)

Then $XBTD$ is a parallelogram (1 pair of $\parallel$ equal sides) (1)

Since XYD is a parallelogram

XY & BD bisect each other (1)



$$5x^2 = 20^2$$

$$x^2 = 80$$

$$x = 4\sqrt{5}$$

$\therefore$ Harry walked $4\sqrt{5}$ km

(ii) $\tan \theta = \frac{1}{2}$

$$\theta = 26^\circ 34'$$

$$\text{Bearings} = 360 - (135 + 26^\circ 34')$$

$$= 198^\circ 26'$$

10a) $\text{LHS} = \frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta}$

$$= \frac{\cos \theta + \cos \theta \sin \theta - \cos \theta + \cos \theta \sin \theta}{1 - \sin^2 \theta}$$

$$= \frac{2 \sin \theta \cos \theta}{\cos^2 \theta}$$

$$= 2 \tan \theta$$

b) $\therefore$ solve

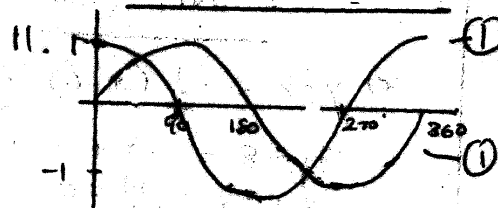
$$2 \tan \theta = \cot \theta$$

$$2 \tan \theta = \frac{1}{\tan \theta}$$

$$\therefore \tan^2 \theta = \frac{1}{2}$$

$$\tan \theta = \pm \frac{1}{\sqrt{2}}$$

$$\therefore \theta = 35^\circ 16', 144^\circ 44', 215^\circ 16', 324^\circ 44'$$



$$\sin x = \cos x$$

$$\text{When } \tan x = 1 \text{ i.e. } x = 45^\circ, 225^\circ$$

$$\sin x \geq \cos x$$

$$45^\circ \leq x \leq 225^\circ$$

12. Solve for $0 \leq \theta \leq 360^\circ$

$$2 \tan^2 \theta - 5 \sec \theta + 5 = 0$$

$$2(\sec^2 \theta - 1) - 5 \sec \theta + 5 = 0$$

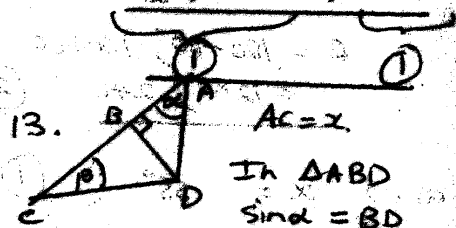
$$2 \sec^2 \theta - 5 \sec \theta + 3 = 0$$

$$(2 \sec \theta - 3)(\sec \theta - 1) = 0$$

$$\sec \theta = \frac{3}{2} \quad \sec \theta = 1$$

$$\therefore \cos \theta = \frac{2}{3} \quad \cos \theta = 1$$

$$\theta = 48^\circ 11', 311^\circ 49', 0, 360^\circ$$



In $\triangle ABD$

$$\sin \alpha = \frac{BD}{AD}$$

$$\therefore AD = \frac{BD}{\sin \alpha}$$

In $\triangle ACD$ $\frac{AD}{\sin \beta} = \frac{x}{\sin(180^\circ)}$

$$\therefore AD = \frac{x \sin \beta}{\sin(\alpha + \beta)}$$

$$\frac{BD}{\sin \alpha} = \frac{x \sin \beta}{\sin(\alpha + \beta)}$$

$$\therefore BD = \frac{x \sin \alpha \sin \beta}{\sin(\alpha + \beta)}$$